

Zhanshan Zhang

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WORK EXPERIENCE

AWS Redshift

East Palo Alto, CA

Software Engineer, Redshift Control Plane Infra

Since July, 2025

- Developed **GenAI-powered Patch Blocker Infra** for triage and RCA analysis of failed tests. Built automated, **rich GenAI context** collection aggregating **Redshift** cluster logs, test artifacts, DB instance workflows (**AWS SWF**), similarity SIMs. Used **Amazon Bedrock Agentcore** to enable an **agentic workflow**. Analyzed over **15,000** patch blockers within months, achieving a **65% RCA match score**. Achieved a **30-minute efficiency gain** per ticket and **95%** dev satisfaction.
- Built **Monitoring Service** for GenAI Patch Blocker Infra using **Python Lambda** and **CloudWatch**, with **real-time** coverage and evaluation metrics, achieving a **98.9%** coverage rate. Deployed using **AWS CDK**.
- Completed a **POC** and **Design Doc** to validate **End-to-end Test Infra refactoring**. Restructured integrations across Redshift DB Test Infra, Patch Blocker Infra, Ticketing Infra. Results indicate **1-3 days reduction** in cluster retention time, **response time reduction from 30 to 5 minutes**, and great **reduction in duplicate infra**.
- Developed a patch blocker **auto-routing** module, getting **70% accuracy** for CP SIMs—**20% higher** than manual assignment.
- Build a **Kendra** index connected to **S3** service, establishing foundational infra for a PB ticket **Knowledge Base** and **chatbot**, enabling easy integration and ticket similarity search capabilities.
- Operational success: Completed host migration (reducing related sev2 to **zero**); Developed a snapshot quota monitoring module; Participated in **4 on-call rotations**, resolving **30+ Sev-2** incidents and **100+** total tickets.
- AWS Services Experience**: Lambda, S3, Redshift, Dynamodb, CloudWatch, Bedrock, AgentCore, Kendra, EventBridge, Step Function, SWF, QuickSuite

PDD Group (Operates Temu & Pinduoduo)

Shanghai, China

Backend Engineer Intern (Ads & Recommendation Engine Group)

July - September, 2024

- Researched the recommendation engine architecture (multi-channel recall, preranking, ranking, filtering, reranking) and scoring system (GMV, CTR, CVR).
- Responsible for several features. Achieved income growth: New iPhone coupon/tags: **GMV +0.12%**, **Revenue +0.09%**. Homepage video autoplay: **CTR + 0.18%**, **GMV +0.14%**. Filtering high-refund goods: **Refund rate** reduced by 30%.
- Added 2 operators to engine's DAG graph for video fetching and preranking-score-based video recommendation.
- Used **Java Future** & **Guava** to develop **asynchronous APIs** for batch **RPC** calls, reducing API response times by 19%.
- Collaborated with frontend/database engineers and integrated backend services (Items, coupons, tags) after engine's reranking phase for iPhone promotion feature, and contributed to **API documentation** for seamless communication.
- Enhanced code quality & team collaboration through active participation in **GitLab** reviews.

Fengxiang Interactive

Xi'an, China

Online Forum Developer

July - August, 2022

- Developed a feature-rich online forum using **Java**, **SpringBoot**. Designed **RESTful APIs** for posting, liking, and following.
- Managed user authentication and authorization using **Spring Security** and interceptors, reducing malicious activity by 67%.
- Employed **MySQL** for data storage with transaction isolation control. Utilized **Redis for caching** & high-frequency data storage, improving performance by over 200% in unit tests.
- Integrated **Kafka** for real-time messaging and **Caffeine** for **server-side caching**, achieving 15~20 times performance boost in **multi-threaded simulations**. Developed comprehensive **JUnit** tests, increasing test coverage by 30%.

TECHNICAL SKILLS

Languages: Java, Python, CDK, JavaScript/TypeScript, C, C++, HTML, CSS

Frameworks: SpringBoot, Node.js, Android Studio, Flask, Angular, MyBatis

Database & DevOps: MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, Kafka, Docker, Pytest, JUnit, Selenium, Linux

Cloud & Version Control: AWS, Git, Github, CRUX

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

August 2023 – May 2025

- Coursework: Analysis of Algorithms, Web technologies, Databases, NLP

National University of Singapore

Singapore

- Exchange Student, A-level Graduation Project

January 2023 – May 2023

Xi'an Jiaotong University (C9 Top League)

Xi'an, China

Bachelor of Science in Computer Science | GPA 3.98 (Top 2%)

September 2019 - December 2022

- Honors and Awards: 2nd Prize Scholarship, 2nd Prize of 13th "Blue Bridge" Software Comp (Python)